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SUPERSEDING
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10 July 1964

MILITARY SPECIFICATION

COPPER ALLOY NUMBER 175 (COPPER-COBALT-BERYLLIUM) BAR AND ROD

1. SCOPE

1.1 Scope. This specification covers one alloy of copper-cobalt-beryllium bar and rod.

1.2 Classification.

1.2.1 Tempers. Copper alloy number 175 bar and rod shall be furnished in the tempers listed below, as specified. "T" following the temper designation denotes material after precipitation hardening treatment.

A - Cold rolled or drawn and solution heat treated.
H - Cold rolled or drawn, solution heat treated and rolled or drawn hard.
AT - Cold rolled or drawn, solution heat treated and precipitation hardened.
HT - Solution annealed, rolled hard or drawn and precipitation hardened.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY
MIL-C-3993 - Copper and Copper Alloy Base Mill Products, Packaging of.

/FSC 9530/

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STANDARDS

FEDERAL

Fed. Std. No. 146 - Tolerances for Copper and Copper Base Alloy Mill Products
Fed. Test Method Std. No. 151 - Metals; Test Methods
Fed. Std. No. 185 - Identification Marking of Copper and Copper Base Alloy Mill Products

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARD

E 8 - Tension Testing of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample shall meet all the requirements of this specification.

3.2 Chemical composition. The material shall conform to the chemical requirements specified in table I.

Table I. Chemical composition

Element	Percent
Beryllium	0.40-0.7
Cobalt	2.4-2.7
Copper plus beryllium plus cobalt	99.5 min.

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3.2.1 Analysis shall be made regularly only for the elements specifically mentioned in table I. If however, the presence of other elements is suspected, or indicated in the course of routine analysis, further analysis shall be made to determine that the total of other elements is not in excess of the limits specified.

3.3 Mechanical properties. Material shall conform to the mechanical properties shown in table II for the respective tempers.

Table II. Mechanical properties

Temper	Tensile strength, psi	Yield strength, 0.2 percent offset min., psi	Elongation in 2 in., min., percent
A	55,000 max.	----------	20
H	65,000-80,000	----------	10
AT	100,000 min.	75,000	10
HT	110,000 min.	95,000	8

3.3.1 Samples of material supplied in the A or H temper shall be capable of meeting the requirements of the AT or HT temper when precipitation hardened according to table III. Samples of material supplied in the A or H temper shall be precipitation hardened in accordance with table III and shall conform to the mechanical property requirements of table II for the corresponding precipitation hardened temper. Samples of material up to 3/4 inch, inclusive, in the least dimension shall be precipitation hardened in full cross section. Samples of bar or rod greater than 3/4 inch in the least dimension may be machined to test-specimen dimensions before the precipitation-hardening treatment.

Table III. Precipitation hardening treatment

Temper	Number of hours at 900° ± 5°F 1/
A	3
H	2

1/ All parts of the charge shall be up to the specified temperature before beginning the time record. No distinction shall be made for time at heat between liquid bath or air-furnace type of heat treatment.

3.3.2 Samples supplied in the AT or HT temper shall have been precipitation hardened according to table III.

3.4 Dimensional tolerances. The following references of Fed. Std. No. 146 shall apply.

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Form	Dimension	Reference
Bar	Thickness	2b (1)
	Width	2b (2)
	Length	2b (3), 2b (4)
	Straightness	2b (5)
	Standard edge contours	2b (6)
Rod	Diameter	11b (1)
	Length	11b (2), 11b (3)
	Straightness	11b (4)

3.5 Identification marking. When specified in the contract or order, product identification marking shall be in accordance with Federal Standard No. 185 (see 6.2 and 6.3).

3.6 Workmanship. Bar and rod shall be uniform in quality, and temper; shall be clean, sound and smooth; and shall be free from injurious defects including foreign material, pipes, slivers, laps, cracks, twists, seams, scale, damaged ends, corners, or edges which due to their nature, degree or extent, detrimentally affect the suitability of the material for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. Unless otherwise specified, (see 6.2), the preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of copper-cobalt-beryllium alloy. The preproduction sample shall be subjected to all examinations and tests specified herein. Preproduction samples which do not meet all the requirements of this specification shall be rejected. Before submitting another preproduction sample, full particulars concerning previous rejection and the action taken to correct the defects shall be made available to the contracting officer. Subsequent quantities will not be considered for acceptance until approval of the preproduction sample has been obtained.

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4.3 Lot. Unless otherwise specified in the contract or order, a lot shall consist of 1,000 pounds or fraction thereof, of material of the same form, temper, and size.

4.4 Sampling.

4.4.1 Chemical analysis. A sample for analysis shall be taken from each lot. At least four pieces representative of the lot shall be selected. From the pieces selected, equal amounts of clean clippings, drillings, or millings shall be obtained, combined and thoroughly mixed into a sample of not less than 5 ounces. If the producer's method of composition control is acceptable, samples for check analysis may be waived at the discretion of the purchaser.

4.4.2 Tension tests. Two tension test samples shall be selected from each lot for determination of mechanical properties of the material before precipitation hardening. Two additional tension-test samples shall be selected from each lot for determination of mechanical properties after precipitation hardening. Each specimen shall be selected from a different piece unless the number of pieces in the lot is less than the required number of specimens. In this case, at least one specimen shall be selected for determination of tensile properties after precipitation hardening (see above), in addition to one for as-supplied properties.

4.4.3 Sampling for visual and dimensional examination. If the weight of each piece is more than 150 pounds, every piece shall be examined. If the weight of each piece is 150 pounds or less, a representative sample of material shall be selected from each lot in accordance with Inspection Level III, MIL-STD-105, acceptable quality level (AQL) of 1.5 percent, and shall be visually examined and measured as specified in 4.5.2. The samples selected for dimensional examination may be the same as those selected for visual examination but shall be evaluated separately.

4.5 Examination.

4.5.1 Pieces weighing over 150 pounds. Each piece in the lot shall be visually examined to determine compliance with the requirements for identification marking (see 3.5 and 6.3) and workmanship (see 3.6) and shall be measured for compliance with the dimensional requirements (see 3.4) of this specification. Straightness of each piece shall be determined as specified in 4.5.3.

4.5.2 Pieces weighing 150 pounds and under. Each sample unit selected in accordance with 4.4.3 shall be visually examined and measured to determine compliance with the requirements for dimensions (see 3.4), identification marking (see 3.5) and workmanship (see 3.6). Straightness of each sample unit shall be determined as specified in 4.5.3.

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4.5.3 Straightness. Straightness shall be determined by placing the piece on a level surface so that the arc or departure from straightness is horizonal. The maximum depth of arc shall be measured to the nearest 1/32 inch by means of a straightedge and a steel scale.

4.5.4 Preparation for shipment. Examination of the packing and marking for shipment shall be made for conformance to the requirements of section 5.

4.6 Tests.

4.6.1 Tension test specimens. Whenever practicable, bar and rod shall be tested in full section. Specimens for tensile testing shall be prepared in accordance with ASTM E 8.

4.6.2 Test procedures.

4.6.2.1 Chemical analysis. The samples selected in accordance with 4.4.1 shall be analyzed in accordance with method 111 or method 112 of Fed. Test Method Std. No. 151 to determine conformance to 3.2. A single analysis of a composite sample may be made. In case of dispute, the analysis by the wet method (method 111) shall be the basis for acceptance.

4.6.2.2 Tension tests. All tension tests shall be conducted in accordance with ASTM E 8.

4.7 Rejection.

4.7.1 Examination defects. Any sample unit having one or more defects shall be rejected. If the number of nonconforming sample units in the sample exceeds the acceptance number specified in 4.4.3 for that sample size, the entire lot shall be rejected, subject to the provisions of the section on "Acceptance and Rejection" of MIL-STD-105 using tightened inspection.

4.7.2 Test failures. A lot shall be rejected for failure to meet any of the test requirements when tested in accordance with 4.6, subject to the rejection and retest provisions of Fed. Test Method Std. No. 151.

4.7.2.1 Failure due to inadequate heat treatment response. When failure of any lot of bars or rods to meet the requirements of this specification is due to inadequate heat treatment response, the material may be reheat-treated and resubmitted for tests. Only two such reheat-treatments shall be permitted, after which, if the specimen fails, no retests shall be permitted.

5. PREPARATION FOR DELIVERY.

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5.1 Packing (see 6.2). The material shall be packed in accordance with Level A, B, or C of MIL-C-3993 as specified.

5.2 Marking. In addition to any special marking required in the contract or order (see 6.2), marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The bar and rod covered by this specification are intended for use in the manufacture of parts requiring good strength and wear resistance, high conductivity and application at high operating temperatures.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Form, size and temper required (see 1.2)

(c) Whether a preproduction sample is required (see 3.1 and 4.2)

(d) Lengths, whether specific or stock lengths with or without ends (see 3.4)

(e) When product identification marking in accordance with Fed. Std. No. 185 is required (see 3.5)

(f) When a lot of other than 1,000 pounds is required (see 4.3)

(g) Whether material is to be packed by Level A, B, or C (see 5.1)

(h) Maximum gross weight of container (see 5.1)

(i) Special marking, if required (see 5.2)

6.3 The requirements for product identification marking (see 3.5) and for packing (see 5.1) and marking (see 5.2) for shipment specified herein apply to direct shipment for Government activities and apply also, when specified, to contracts or orders between the manufacturer and the Government prime contractor.

6.4 Hardness and electrical conductivity. The values shown in table IV for hardness and electrical conductivity of rod and bar are given for information only.

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Table IV. Hardness and electrical conductivity

Temper	Rockwell Hardness	Electrical conductivity IACS, minimum percent
A	B20-50	20
H	B60-75	20
AT	B92 min.	45
HT	B95 min.	48

6.5 Definition of forms.

6.5.1 Bar. A bar is a solid rectangular section, or one with two plane parallel surfaces and round or other simple regularly shaped finished edges, up to and including 12 inches in width and over 0.188 inch in thickness furnished in straight lengths or in rolls and with finished edges, either rolled, drawn or extruded.

6.5.2 Rod. A rod is a round, regular hexagonal or regular octagonal solid section furnished in straight lengths. Note: A regular hexagonal or a regular octagonal rod is a solid section having equal sides and equal angles.

Custodian:
Army - MR

Preparing activity:
Army - MR

Review activities:
Army - MR, MU, MI
DSA - IS

Project No. 9530-A004

User activities:
Army - ME, EL, AT, WC

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